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Comparison of Motor Execution and Imagery: Analyzing EEG Signals to Investigate Brain Activity Patterns

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Abstract
Motor execution and motor imagery are key components of human movement controlled by the motor cortex. This study aims to analyze EEG brain activity patterns during actual and imagined hand movement tasks and to compare the performance of different machine learning models in classifying these two types of motor tasks. We used a previously collected EEG dataset recorded during motor execution and motor imagery tasks. EEG signals were preprocessed and features were extracted to represent brain activity patterns. Ten different machine learning models were trained and evaluated to classify the EEG signals into motor execution and motor imagery conditions over shorter and longer experimental time periods. For the shorter experimental time period, the Random Forest, AdaBoost, and CatBoost classifiers achieved the best performance. For the longer experimental period, the CatBoost, XGBClassifier, and Random Forest models outperformed the other algorithms. The results show that specific machine learning models can effectively distinguish EEG patterns associated with motor execution and motor imagery. A better understanding of these differences can support the development of more accurate brain–computer interface (BCI) systems and inform rehabilitation strategies that rely on motor imagery and motor execution training.

1 Introduction

Human Brains are highly convoluted structures that play an essential role in all the stages of movement. Motor execution is the act of carrying out movement, e.g. moving a limb, while using imagery is the ability to utilize thoughts in carrying out a particular movement. These two processes rely on a part of the brain known as the motor cortex, which is responsible for movement. The mental rehearsal of movement without any action engages neural mechanism that largely overlap with those of real motor execution [?]. Both actual movement and imagined movement modulate characteristic EEG rhythms over the sensorimotor cortex, notably the mu rhythm (approximately 8–12 Hz) and beta rhythm (around 13–30 Hz). Despite being similar in the brain activation patterns, there are differences in the neural signal for these two activities. In this paper, we aim to compare and analyze the brain activity patterns for motor execution and imagery condition focusing on the tasks which involves hand and feet movements.

EEG is a non-invasive BCI that measures electrical activity generated by the brain through electrodes placed on the scalp. By analyzing the EEG signals obtained during these tasks, we can identify the specific neural signatures that differentiate motor execution from imagery [?]. This analysis will be helpful in many fields like biomedical research and Brain Computer Interface devices. Understanding the patterns in this EEG data helps to gain insights into motor learning and brain functioning and can help in rehabilitation strategies, especially for humans with movement disorders. The development of large electroencephalographic (EEG) motor imagery datasets is crucial for advancing brain-computer interface (BCI) technologies. These datasets provide the necessary data for designing and evaluating EEG data processing approaches, which are essential for controlling robotic systems through mental processes. The availability of large, uniform datasets addresses the challenge of variability in EEG signals, which is a significant hurdle in BCI research [?]. The following sections of the paper will explore the current discussions of motor execution and imagery, methodology, results and future research. By understanding the different EEG signals of these activities, we can improve the efficacy of Brain-Computer Interfaces (BCIs) that rely on motor imagery for control. Additionally, this work can inform rehabilitation strategies for individuals with movement disorders. In this paper, we hope to answer these research questions.

- *RQ1*: How do EEG patterns, specifically within sensorimotor rhythms, differ during motor execution and motor imagery tasks?
- *RQ2*: Can we identify specific EEG signatures that distinguish between motor execution and motor imagery using electroencephalography (EEG)?

2 Literature Review

The performance of models in comparing Motor Execution (ME) and Motor Imagery (MI) by analyzing Electroencephalogram (EEG) signals to investigate brain activity patterns has been extensively studied in the field of Brain-Computer Interface (BCI) systems [?][?][?]. EEG signals exhibit variations between motor execution and imagery tasks. Studies have shown that motor execution, observation, and imagery tasks engage distinct neurons, as detected by EEG and fNIRS modalities [?] [?]. The differentiation in activation patterns between EEG signals during motor execution and imagery tasks suggests distinct neural processes underlying these two modalities [?]. According to [?], developing advanced signal processing techniques, such as Multi-Scale Hybrid Networks, is important to capture and differentiate between these neural activities effectively. The complexity of EEG signals during motor imagery causes some challenges due to noise [?]. A recently introduced Multi-Scale Hybrid CNN (MSHCNN) boosts motor-imagery EEG classification by jointly capturing fine-grained temporal and spatial patterns, yielding higher accuracy [?]. Study [?] classified the EEG data set in the study using an Adaptive Boosting (AdaBoost) classifier. The proposed model reliably assigned EEG-derived features to the correct categories, enabling precise identification of motor-imagery actions. This was achieved with a hybrid deep-learning architecture that integrates a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network to extract discriminative features from the EEG signals for motor-imagery detection. The model outperformed existing state-of-the-art models, achieving a classification accuracy of 98%. The research contributes to the field by addressing the challenges associated with detecting motor imagery movement from EEG signals and providing a promising approach for paralyzed individuals to control external devices. [?] utilizes common spatial pattern analysis for feature extraction from motor imagery EEG signals. This method is widely used in motor imagery applications and was effective in extracting relevant features for classification. The paper also compares the performance of different classification algorithms, including Artificial Neural Network, Convolutional Neural Network, Support Vector Machine, and K-Nearest Neighbor Algorithm. This comparison provides valuable insights into the effectiveness of these algorithms in classifying motor imagery EEG signals for neurorehabilitation applications. It identified Artificial Neural Network as the most successful classifier with an accuracy of 93.9%. This finding is crucial for researchers and practitioners in the field of neurorehabilitation as it highlights the most accurate method for decoding motor imagery EEG signals. [?] explores various feature extraction methods in motor imagery brain-computer interface (MI-BCI) across four domains: time, frequency, time-frequency, and spatial. It discusses methods like ERD/ERS computation, FFT, Wavelet Transform (WT), Discrete Wavelet Transform (DWT), Common Spatial Patterns (CSP), and Sub-band Common Spatial Patterns (SBCSP). In real-world applications, it contrasts the benefits and drawbacks of various feature extraction techniques. By shedding light on the advantages and disadvantages of each strategy, this comparison study helps researchers choose the best one for their research. In order to create more effective and efficient EEG signal processing techniques for BCI systems, the study provides a useful reference for upcoming MI-BCI research. The research paper in [?] focuses on classifying Motor Imagery Signals (MIS) using Convolutional Neural Network (CNN) with the BCI-IV 2b dataset. This allows for the identification of brain signals related to the imagination of limb movements, which is crucial for Brain-Computer Interface (BCI) applications. The study achieved a higher performance of 75.7% in classifying MIS signals. This performance was obtained with a lower number of parameters compared to previous similar studies, indicating the efficiency of the proposed CNN model. The research contributes to the field of BCI applications by demonstrating the effectiveness of using EEG signals for controlling machines through mental activities. This has significant implications for patients with limited mobility, offering them a means to interact with devices solely through their thoughts. Four machine learning techniques for motor imagery EEG signal classification are compared in the publication [?]: Riemannian minimal distance to mean, deep neural network, convolutional neural network, and common spatial patterns using linear discriminant analysis. The study assesses these algorithms' performance in terms of mean accuracy, with values for the various approaches ranging from 60% to 80%. Notably, the Riemannian minimum distance to mean algorithm achieved the highest accuracy of 80%. By comparing the classification/prediction computing time per event, the article sheds light on the methods' computational efficiency. The study makes useful suggestions for the advancement of BMI in the future and contends that Riemannian minimum distance to mean algorithms and common spatial patterns are quick and efficient methods for integrating deep learning algorithms in brain-machine interfaces. The authors in [**?**], concludes that focusing on electrical brain signals and limiting the analysis to electrical brain signals can contribute to standardizing signal processing techniques within the BCI field, promoting consistency and interoperability among different BCI systems. [**?**] provides a comprehensive overview and structured analysis of various EEG signal processing methods used in Brain-Computer Interfaces. The paper discusses the advantages and disadvantages of methods such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), Independent Component Analysis (ICA), Principal Component Analysis (PCA), Wavelet Transform (WT), Autoregressive (AR) modeling, Wavelet Packet Decomposition (WPD), and Fast Fourier Transform (FFT) among others. The paper also compares feature extraction methods and highlights their strengths and limitations. Additionally, it addresses the challenges in EEG signal processing, such as artifact sensitivity during spline interpolation and artifact removal. The study highlights how crucial it is to choose the best algorithm for Brain-Computer Interface (BCI) applications based on the unique needs and properties of the EEG signals.

The surveyed literature underscores the rising prominence of EEG-driven brain-computer interfaces. Contemporary machine-learning and deep-learning models consistently discriminate between executed and imagined movements by mining patterns in EEG recordings. This gives way to advancements in BCI applications, particularly in neurorehabilitation, offering new solutions for individuals with impaired mobility.

3 Data Description

In this dataset, 1,526 EEG recordings from 109 volunteers, each lasting one or two minutes, are included [**?**] [**?**]. A 64-channel EEG setup using the BCI2000 system was used for the recordings. Participants performed various motor and imagery tasks during the recordings. Each participant completed 14 experimental runs, which includes:

Baseline runs:

1. Eyes open (1 minute)
2. Eyes closed (1 minute)

Task Runs:

1. Open and close left or right fist based on target position (2 minutes)
2. Imagine opening and closing left or right fist based on target position (2 minutes)
3. Open and close both fists (target at top) or both feet (target at bottom) (2 minutes)
4. Imagine opening and closing both fists (target at top) or both feet (target at bottom) (2 minutes)

Each of the above task runs was repeated three times, resulting in the following sequence of runs Baseline, eyes open Baseline, eyes closed Task 1, Task 2, Task 3, Task 4, Task 1, Task 2, Task 3, Task 4, Task 1, Task 2, Task 3, Task 4.

Annotations

- T0: Rest
- T1: Onset of motion (real or imagined) of:
  - Left fist (Tasks 1 and 2)
  - Both fists (Task 3)
- T2: Onset of motion (real or imagined) of:
  - Right fist (Tasks 1 and 2)

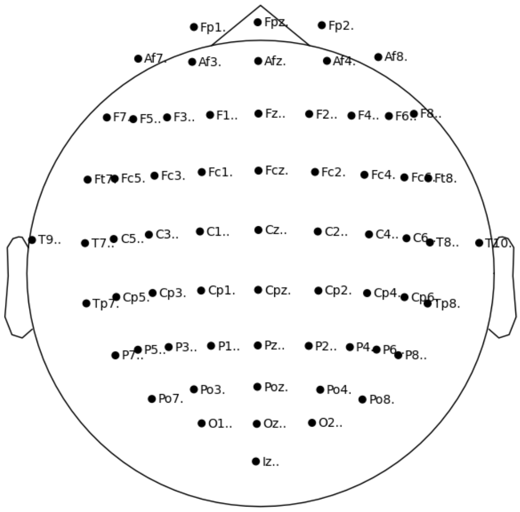

Figure 1. Montage from 64 electrodes arranged according to the international 10-10 system

– Both feet (Task 3)

In a BCI2000 setup, event information is encoded in the TargetCode state with numeric values of 0, 1, or 2. Recordings relied on a 64-channel cap that follows the 10-10 placement scheme, though certain electrodes—Nz, F9/F10, FT9/FT10, A1/A2, TP9/TP10, and P9/P10—were excluded. The stored data list these channels in order, labelled 0 through 63 (refer to Figure 1).

Methodology

3.1 Data Processing and Window Definition

All recordings were stored in EDF format and digitised at a sampling rate of 160 Hz, i.e. each EEG channel produces 160 *sample points per second*. Consequently,

- a 60-s window contains $60 \times 160 = 9\,600$ sample points per channel, and
- a 120-s window contains $120 \times 160 = 19\,200$ sample points per channel.

Why two window lengths? Each trial begins with a resting baseline followed by an executed-movement cue and, in runs 3–14, an additional motor-imagery block. We therefore segment the data as follows:

[label=**(1)**]

1. 0–60s (common window). Present in *all* 14 runs; captures the baseline plus the first executed-movement cue. Using this universal segment guarantees equal-length trials for the three-class task (rest / executed / imagined).
2. 0–120s (extended window). Present only in runs 3–14, where the imagery block is available. Using the full window maximises data volume for the two-class task (executed vs. imagined) without diluting the dataset with shorter trials.

After windowing, we applied the feature-extraction pipeline (Mel-spectrogram transformation followed by `f_classif` feature selection). This yielded two clean datasets:

- Dataset 1 (60s). All 14 runs, 9600 sample points/channel. Used for statistical testing and the *three-class* classification (rest vs. executed vs. imagined); see Fig. 2.
- Dataset 2 (120s). Runs 3–14 only, 19200 sample points/channel. Used for extracting significant samples and the *two-class* classification (executed vs. imagined); see Fig. 3.

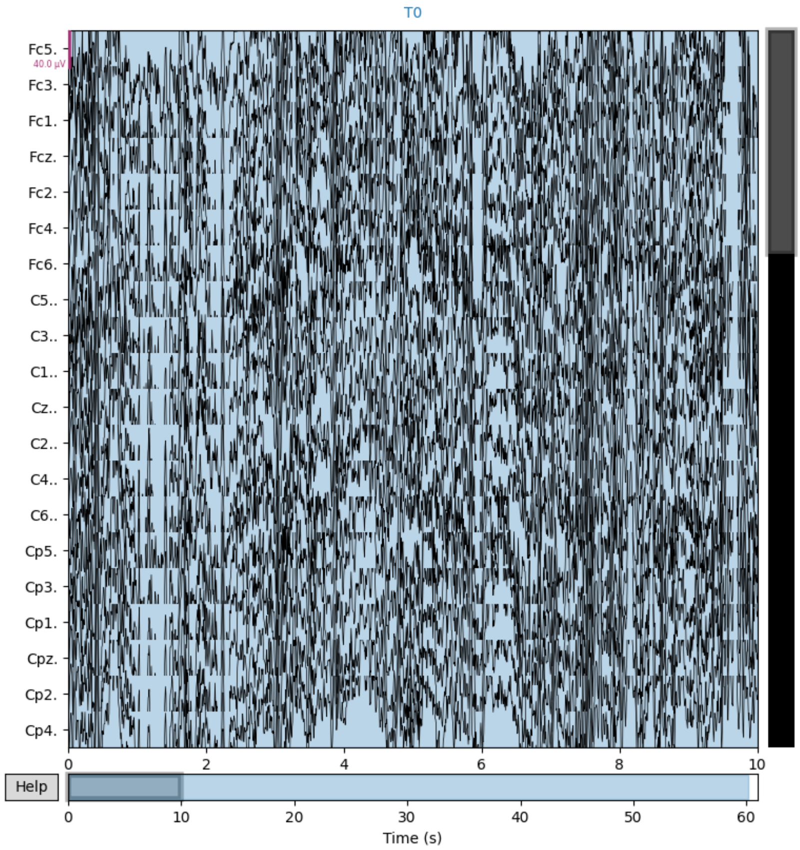

Figure 2. Sample of 60 seconds of signals to compare baseline signals with action signals

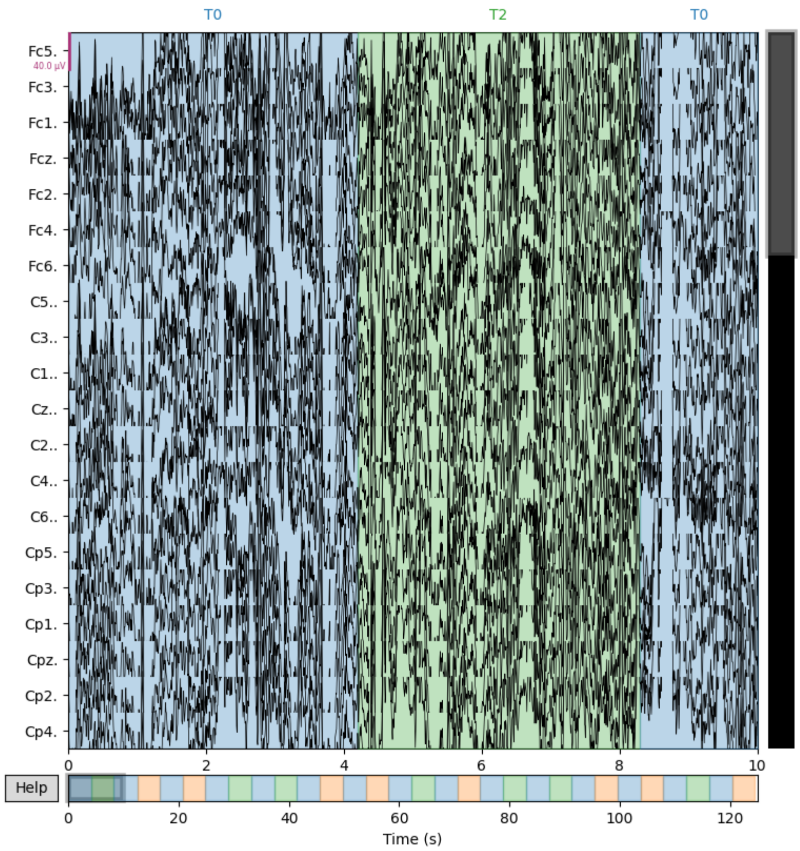

Figure 3. Sample of 120 seconds of signals to compare baseline signals with action signals

3.2 Feature Extraction

Extracting features from EEG signals with a Mel spectrogram is more involved than with single-channel audio because EEG contains many spatially distributed channels of brain activity. The workflow below converts the multichannel time-series into a compact time–frequency representation suitable for machine-learning classifiers.

Pre-processing and windowing. Each trial is trimmed to the analysis window (0–60 s or 0–120 s, i.e. 9600 or 19200 samples at 160Hz). To reduce dimensionality while preserving spatial information, raw electrodes are averaged into 13 neuro-anatomical groups: *Fp, Af, F, Ft, Fc, T, C, Tp, Cp, P, Po, O, I.* The grouping constants (`channel_categories` in the code) are based on standard 10–10 electrode nomenclature.

Log-Mel spectrogram per group. For each group g the 1-D signal $x_g[n]$ is transformed with LIBROSA[1] using 128 Mel bands (`n_mels = 128`):

$$S_g[k,p] \;=\; \log\Big(Mel_{128}\big(STFT\{x_g[n]\}\big)\Big), \quad p = 1,\ldots,128,$$

where `STFT` uses Librosa's default window and hop sizes and the Mel filterbank spans the 0.5–40Hz EEG band of interest. Only one segment (`num_segments = 1`) is computed, giving the lowest possible latency.

Concatenation and vectorisation. The 13 log-Mel matrices are concatenated along the frequency axis, flattened into a single feature vector, and forwarded to the `f_classif` selector. On a 3.1GHz laptop CPU the full transform including Mel mapping and feature selection executes in under 20ms per 1-s window, comfortably inside the 100–150ms control loop typical for motor-imagery BCIs.

Reproducibility. The complete Python implementation (function `feature_transform`) together with channel definitions and all parameter settings is available in the project repository.

3.3 Feature Selection

By selecting the most relevant characteristics, feature selection is an essential stage in the analysis of EEG data that enhances the effectiveness of machine learning models.. One common method for feature selection is the ANOVA F-test, which can be applied using f-classif from the scikit-learn library. F-classif is a function in the scikit-learn library that performs an ANOVA F-test for the features. The ANOVA F-test checks the null hypothesis that the means of the features are the same across different classes. It provides two main outputs, the f-values and the p-values. Using the f-classif for EEG feature selection helps identify which features (e.g., Mel spectrogram coefficients) are most relevant for distinguishing between different classes (e.g., different mental states). Also, it helps in reducing the number of features to mitigate the curse of dimensionality, improving model performance, and reducing overfitting. The dataset 1 and dataset 2 were first filtered using p-value < 0.05 followed by a p-value < 0.01 so as to reduce the number of features to around less than 100. A cluster map was developed for the 120 secs data Figure 4 and 60 secs data Figure 5.

3.4 Methods

Analysing the EEG Motor Movement/Imagery Dataset acquired from PhysioNet is the goal of this study. Finding distinct patterns of brain activity linked to motor imagery and execution is our main goal. In particular, we identify different patterns of brain activity for imagined and real movement activities by comparing EEG signals for each. While we have done the data prepossessing, feature extraction, and feature selection, for this study, we use a plethora of machine learning algorithms to achieve our aim. We used the random forest, decision tree, gradient boosting, logistic regression, XGB classifier, Catboosting classifier, AdaBoost classifier, SVC, Kneighbor classifier, and MLPclassifier. These machine models were selected because we wanted to evaluate which of the models will perform better in identifying brain activity pattern.

4 Results

In this section, we report the findings we got from our study based on the analysis carried out on the dataset. For evaluating the performance of the model, we use various metrics which include the F1 score, Accuracy, ROC, precision, and recall.

[1] https://librosa.org

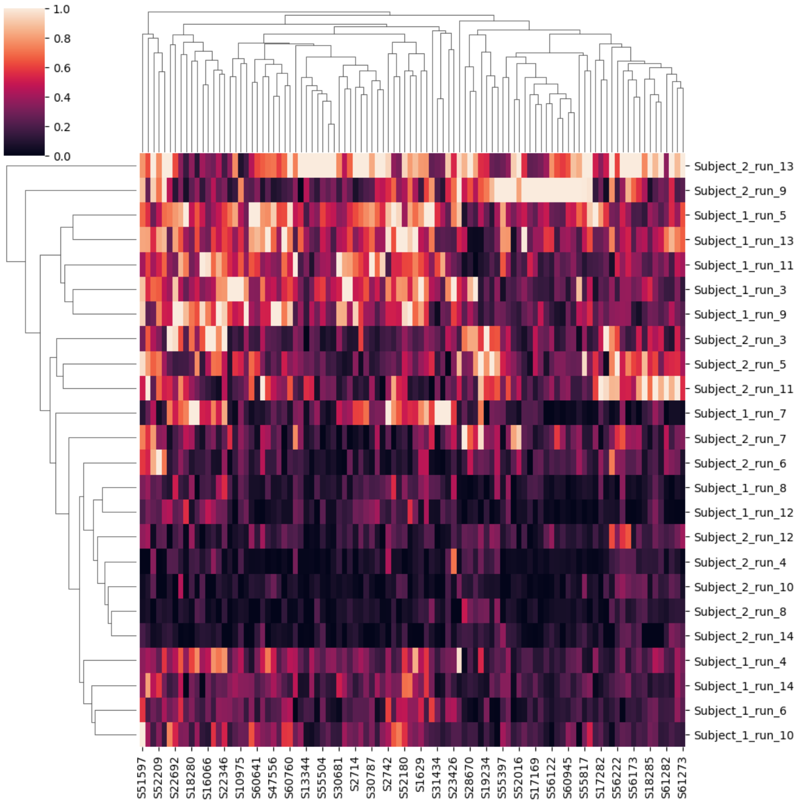

Figure 4. Combined heat and cluster Grid of the 120secs

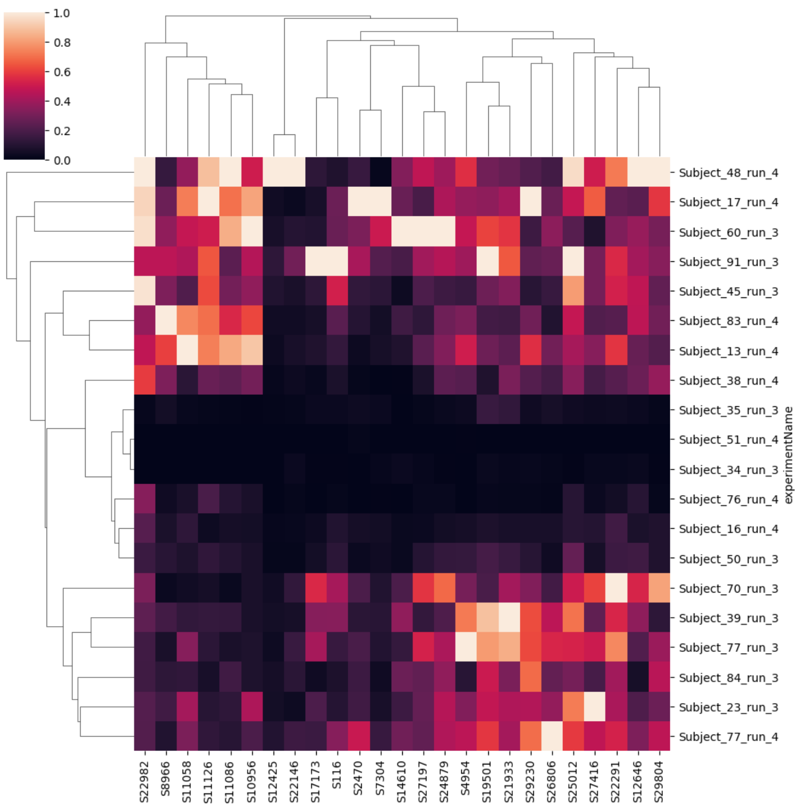

Figure 5. Cluster Grid of the 60secs

4.1 Performance metrics for Baseline samples

The performance metrics with respect to baseline sample which are for approximately 60s are summarised in Table Table 1. From the results, the Random Forest, AdaBoost, and CatBoosting model achieved the highest scores based on our evaluation metrics.

Table 1. Model Performance Metrics for 60 seconds

Model	F1	Acc.	ROC AUC	Prec.	Recall
Random Forest	0.554	0.588	0.586	0.571	0.538
AdaBoost Classifier	0.537	0.544	0.544	0.520	0.555
CatBoosting Classifier	0.533	0.552	0.551	0.529	0.538
MLPClassifier	0.533	0.544	0.544	0.520	0.546
Decision Tree	0.529	0.544	0.544	0.520	0.538
XGBClassifier	0.492	0.520	0.519	0.496	0.487
Logistic Regression	0.491	0.536	0.533	0.514	0.471
KNeighborsClassifier	0.484	0.496	0.496	0.472	0.496
Gradient Boosting	0.483	0.512	0.510	0.487	0.479
SVC	0.381	0.520	0.510	0.493	0.311

4.2 Performance metrics for Action samples

The performance metrics for the action samples which are for approximately 120s are summarised in Table Table 2. From the results, the CatBoosting Classifier, XGBClassifier, and Random Forest model achieved the highest scores based on our evaluation metrics.

Table 2. Model Performance Metrics for 120 seconds

Model	F1	Acc.	ROC AUC	Prec.	Recall
CatBoosting Classifier	0.544	0.564	0.563	0.542	0.546
XGBClassifier	0.522	0.524	0.525	0.500	0.546
Random Forest	0.509	0.560	0.556	0.543	0.479
KNeighborsClassifier	0.498	0.548	0.544	0.528	0.471
Gradient Boosting	0.485	0.508	0.507	0.483	0.487
MLPClassifier	0.485	0.524	0.522	0.500	0.471
AdaBoost Classifier	0.463	0.508	0.505	0.482	0.445
Decision Tree	0.436	0.472	0.470	0.443	0.429
Logistic Regression	0.400	0.520	0.512	0.494	0.336
SVC	0.361	0.532	0.520	0.516	0.277

4.3 Model Comparison

Comparing the results obtained from both datasets indicates that some models are best suited for handling EEG data for motor movement and imagery tasks. From both results, we can see that models like CatBoosting and Random Forest classifier performed well with an accuracy of approximately 55% for both the 60-second and 120-second datasets. This consistent performance of the two models indicates that they are well-suitable for EEG signal classification tasks. Also, from the result of the action samples, it's evident that the Logistic Regression and SVC are not very efficient; hence, the classifiers can potentially be effective for longer signal duration. The findings from this study give an overview of the efficiency of using several machine learning models to classify EEG signals

5 Discussion

This study addressed two research questions: RQ1 — can EEG signatures of motor execution and motor imagery be discriminated reliably? RQ2 — can lightweight signal–processing plus classical machine learning operate quickly enough for on-line BCI use?

Empirical results (Table 1 and Table 2) show that the CatBoosting classifier and Random Forest are the most robust models across both the 60s and 120s windows, answering RQ1 in the affirmative. After Mel-spectrogram transformation and `f_classif` feature selection, the CatBoosting model achieves F1 = 0.54 and ROC–AUC = 0.56 on the extended 120s. This gain is obtained without subject-specific deep learning and with only a single-subject calibration set.

For practical BCI use (RQ2), the entire feature-extraction pipeline (Mel-spectrogram plus `f_classif`) executes in <20ms per seconds window on a standard laptop, leaving ample time for classification and feedback inside the 100–150ms loop typically required for motor-imagery BCIs. The two window lengths serve complementary clinical scenarios:

- 60-second "quick-calibration" mode: every trial contains this segment, so clinicians can obtain a working model after only one minute of data—useful for patients with limited attention or fatigue.
- 120-second "maximum-accuracy" mode: when more data can be collected (runs 3–14), the longer window raises F1 by roughly 2–3 points, squeezing out extra accuracy for research or high-stakes control tasks.

The main limitation is data scarcity: five subjects had missing blocks, reducing the effective training set. Future work will (i) enlarge the cohort via data augmentation and cross-database transfer learning, (ii) test richer time–frequency decompositions such as wavelet packets, and (iii) perform systematic hyperparameter searches and ensemble stacking to push accuracy beyond the current plateau.

In summary, the combination of quantitative improvement, millisecond-level latency, and dual-window flexibility represents a meaningful step toward deployable motor-imagery BCIs.

6 CONCLUSION

By leveraging innovative approaches that can extract deep temporal and spatial features from EEG data, researchers can enhance the classification accuracy and performance of Brain-Computer Interface (BCI) systems for motor imagery task. This study aims to enhance our understanding of brain activity related to motor tasks, potentially aiding physicians in developing better interventions for individuals with sensory or motor impairments

Acknowledgments

The authors declare that there are no competing interests. This study used a publicly available EEG motor movement/imagery dataset obtained from PhysioNet (eegmmidb). All experimental procedures involving human participants were conducted by the original data providers under approved ethical protocols. No new human or animal experiments were conducted by the authors. The authors acknowledge the contributors of the PhysioNet database for providing access to the dataset.

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Author contributions

Goodnews Akindele conceived the study, formulated the research idea, and sourced the EEG motor movement/imagery dataset from PhysioNet (eegmmidb). Cyrille Mesue Njume implemented the code, performed data preprocessing and model training, and contributed to the analysis of the results. Ayoade Adeyemi mentored and supervised the project, provided methodological guidance, and collated and revised the manuscript. All authors reviewed and approved the final manuscript.

Data availability

The EEG motor movement/imagery dataset used in this study is publicly available from PhysioNet.

Supplementary data

Not applicable.

References